

Did You Know?

The term "Game Engine" was first used in mid 1990's especially with reference to 3D FPS games

RenderWare Game Engine

Widely used cross platform game engine RenderWare was used in the creation of the Grand Theft Auto & Burnout franchisees



The renderer passes the information of the polygons that make up the car's bonnet (shiny yellow texture) to the graphics card. Also, renderer discards the polygons that make the car's trunk

first level of God of War III, you're talking of hundreds of thousands of vertices and polygonal surfaces.

Consider a game object such as a car. The renderer doesn't really know what exactly a car will look like in the 3D model, what texture it uses (colour and shades design), how it is animated and where it is located in the game. Also it doesn't know where the camera is located and where it is directed. All this information needs to be provided to the renderer by the game.

The game passes information about the object (a car in this case) to the renderer. The renderer, in turn, looks at the size of the 3D model of the car and the location of the camera and then decides if it is on the screen, behind the observer or completely off the screen. It then discards the polygons of this car model that are not faced by the camera. The renderer sees the texture of the polygon and ensures the graphics card uses the same texture while rendering. The graphics card receives the relevant information from the renderer and displays the graphics.

The creation tool - Editor

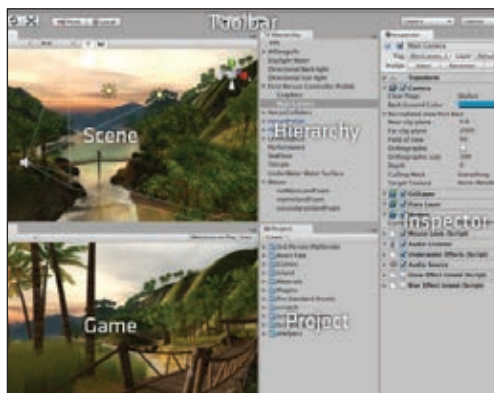
Game engines normally include a creation tool. This is an interface you use to communicate with the game engine. It needs to be flexible to let you drag and drop game objects into the scene, remove them, manage their lighting and save your work. All major game engines let you create projects (games), scenes, import 3D models (game objects) from major 3D editors such as 3ds Max, position them, add levels, adjust and preview lighting, attach physics to your objects, add scripts to them and test run your level before letting it go into the game. The interface of different game engines may be different, but essentially they all do the same thing.

For example, Unity game engine has an interface with a game view and a scene view.

It has an inspector window to adjust the position and size of the game objects and attach scripts. Project and hierarchy windows allow you to manage game objects. Unity also gives you an option to preview the scene in game view using a simple play-pause mechanism.

Lighting and shading

Lighting is an important aspect of game design. All major game engines provide good lighting tools. Vertex lighting, polygon lighting and pixel lighting are some lighting techniques that are widely used.



Interface of Unity game engine

Light sources can be added at desired places in the scene. Many game engines allow you to make light maps, which combine the effect of all directional lights in your scene. This gives you better lighting than vertex lighting and speeds up rendering.

Shaders were introduced to give programmers precise computational control over the final colours displayed on the screen. These days, game engines have built-in Vertex and Pixel shaders. These are programs or routines that help you enhance your graphics by adding complex per-pixel effects. With the above mentioned lighting and shading techniques you can try lighting your game using the various options in Unity.

Character modelling

Character animation and modelling defines how the model looks on screen and how it is animated. By using various design tools available such as 3ds Max and Maya, you can make 3D models and characters. The character models can be imported into the game engines. All game engines support major file formats of 3D modelling tools.

You can import these models into game engines and attach animation scripts to them. Animation scripts are lines of code in a supported language (Java Script in Unity) or the in-built animation scripts like Walk, Run can be used.

Animation requires smooth transitions. In simple terms, if a model is animated to walk, and then run and finally kick, the transitions should not be abrupt. If a number of animations are playing on one character, all of them can be blended and added together to make the final animation run smooth using the "Animation Blending" feature of game engines. Another interesting feature game engines such as Unity provide is animation layering that allows group animations.

Game physics

Similar to the real world physics which all the real objects follow, the objects in the game abide by game physics. Game physics has its own set of rules (sometimes contrary to the opposite of real world physics), to bring in the fun